# Lab 3 – Architectural Style Selection Justification

## Self-Service Coffee Kiosk System

### Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System. The system is organized into a Presentation Layer, Business Layer, and Data Layer, with each layer handling a distinct responsibility.

### Architectural Choice and Reasons

First, Layered Architecture provides clear separation of concerns. The Presentation Layer handles touchscreen interactions, while the Business Layer manages orders, payment processing, and receipt generation. The Data Layer stores menu and pricing information. This separation makes the system easier to understand, maintain, and modify.

Second, Layered Architecture is suitable for this relatively simple kiosk application. The system has limited functionality—customers select a coffee type and size, make a credit-card payment, and receive a receipt—so the additional operational complexity of Microservices Architecture is unnecessary.

### Security Advantage

The layered structure separates the touchscreen interface from the data storage layer. The user interface does not directly access the database; requests are handled through the business layer. This provides controlled access to menu and pricing information and reduces direct exposure of the database.

### Performance Benefit

The separation of responsibilities allows menu and pricing requests to be handled through the appropriate business and data components without unnecessary direct communication from the touchscreen to the database. This organized interaction can reduce unnecessary processing and support efficient kiosk operation.